

Esercizio 1. Let $N \models \mathcal{T}$, such that $N \models \Phi(a)$. Then $\mathcal{T} \cup \{\exists y \phi(x, y)\}$ is consistent.

Proof. By contradiction, let's say it is inconsistent. Then $T \models \neg(\exists x \exists y \phi(x, y))$, so by definition of $\mathcal{T}^+$ we have $\neg(\exists x \Phi(x)) \in \mathcal{T}^+$. Then $N \models \Phi(a) \wedge \neg(\exists x \Phi(x))$, a contradiction.

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